Scott Nguyen

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EDUCATION

Master of Science in Electrical Engineering | Space Systems Engineering **Summer 2026**
University of New Mexico GPA: 3.93/4.00

Master of Science in Aerospace Engineering **Fall 2024**
University of Illinois Urbana-Champaign

Bachelor of Science in Aerospace Engineering **Spring 2022**
Iowa State University

SKILLS

Programming: Python, MATLAB, C/C++, Ruby
GNC / Flight Software: cFS, PX4, MAVSDK, MAVLink, EKF/UKF, Monte Carlo Analysis, Orbit Determination
Simulation / Tools: Simulink, Trick, NumPy, SciPy, Matplotlib, Git, Linux, CI/CD
Applications: ANSYS HFSS, FEKO, TICRA, Blender

WORK EXPERIENCE

Guidance, Navigation & Controls Engineer Intern **January 2026 – May 2026**
NASA Johnson Space Center (JSC)

- Developed and integrated GNC flight software applications within NASA's **core Flight System (cFS)** architecture for the **SPLICE** project.
- Worked on **CAMXIC**, **ECAM**, and **IMU** applications supporting telemetry interfaces, packet routing, and payload integration.
- Built **PX4 + MAVSDK** offboard-control simulations for autonomous translational pointing guidance and trajectory evaluation.
- Developed **Trick-based** simulations, telemetry analysis tools, and automated controller tuning workflows for closed-loop testing.

Student Co-Op, Electronics for Contested Space Group **September 2025 – January 2026**
MIT Lincoln Laboratory

- Developed an optical space domain awareness analysis tool that propagates TLE-based RSO orbits from user-defined sensor locations and applies illumination and elevation constraints to identify observable windows and pointing solutions.
- Modeled RSO detectability and tracking performance for optical space domain awareness using photometric, kinematic, and sensor-noise models to assess detection limits and track duration.
- Designed a communication architecture with radio-frequency hardware and simulation to support lost-satellite reacquisition.

Guidance, Navigation & Controls Engineer Intern **May 2025 – August 2025**
Blue Canyon Technologies

- Executed post-environment testing of **IMU, star trackers, reaction wheels, torque rods, and sun sensors**.
- Diagnosed **Solar Array Drive Assembly (SADA)** timing latency and automated **COSMOS (Ruby)** validation.

Guidance, Navigation & Controls Engineer Intern **January 2025 – April 2025**
Blue Origin

- Designed a hybrid ***Active Disturbance Rejection Control (ADRC)*** + ***Sliding Mode Control (SMC)*** controller; used an extended state observer and a sliding manifold for robust tracking under uncertainty and actuator limits.
- Injected time-domain disturbances and conducted a trade study comparing the hybrid controller to ***PID*** and ***LQR*** on LTI systems; benchmarked rise time, settling time, and overshoot.
- Integrated flight software into ***Simulink*** via ***C S-functions*** to run full closed-loop simulations.

Guidance, Navigation & Controls Engineer Intern **May 2024 – August 2024**
Varda Space Industries

- Conducted trade studies to optimize gravity models for mission requirements and select the best filter (***Extended Kalman Filter (EKF)*** vs. ***Unscented Kalman Filter (UKF)***)
- Created Monte Carlo simulations to perform flight safety analysis and develop reentry criteria for capsule reentry
- Implemented an ***EKF*** for state estimation, optimizing ground station data timing to minimize residuals and enable precise delta-v planning
- Implemented unit testing for the code base and introduced CI/CD pipelines using Bamboo

Guidance, Navigation & Controls Engineer Intern **May 2023 – August 2023**
Space Dynamics Laboratory

- Developed **angles-only orbit determination** pipelines using UKF, range iteration, and least-squares methods for **SSA tracking and characterization**.
- Performed Monte Carlo observability analysis to identify challenging geometries and improve tracking of closely spaced RSOs.
- Implemented and unit-tested a **Lambert-based IOD solver** to support track initialization and catalog maintenance.